ADC
ANALOG TO DIGITAL CONVERTER

Presented by: Group-2
MD Fayed Hasan (2210176143)
MD Shamiur Rahman Sefat (2110276105)
Momin Sarker (2110676137)

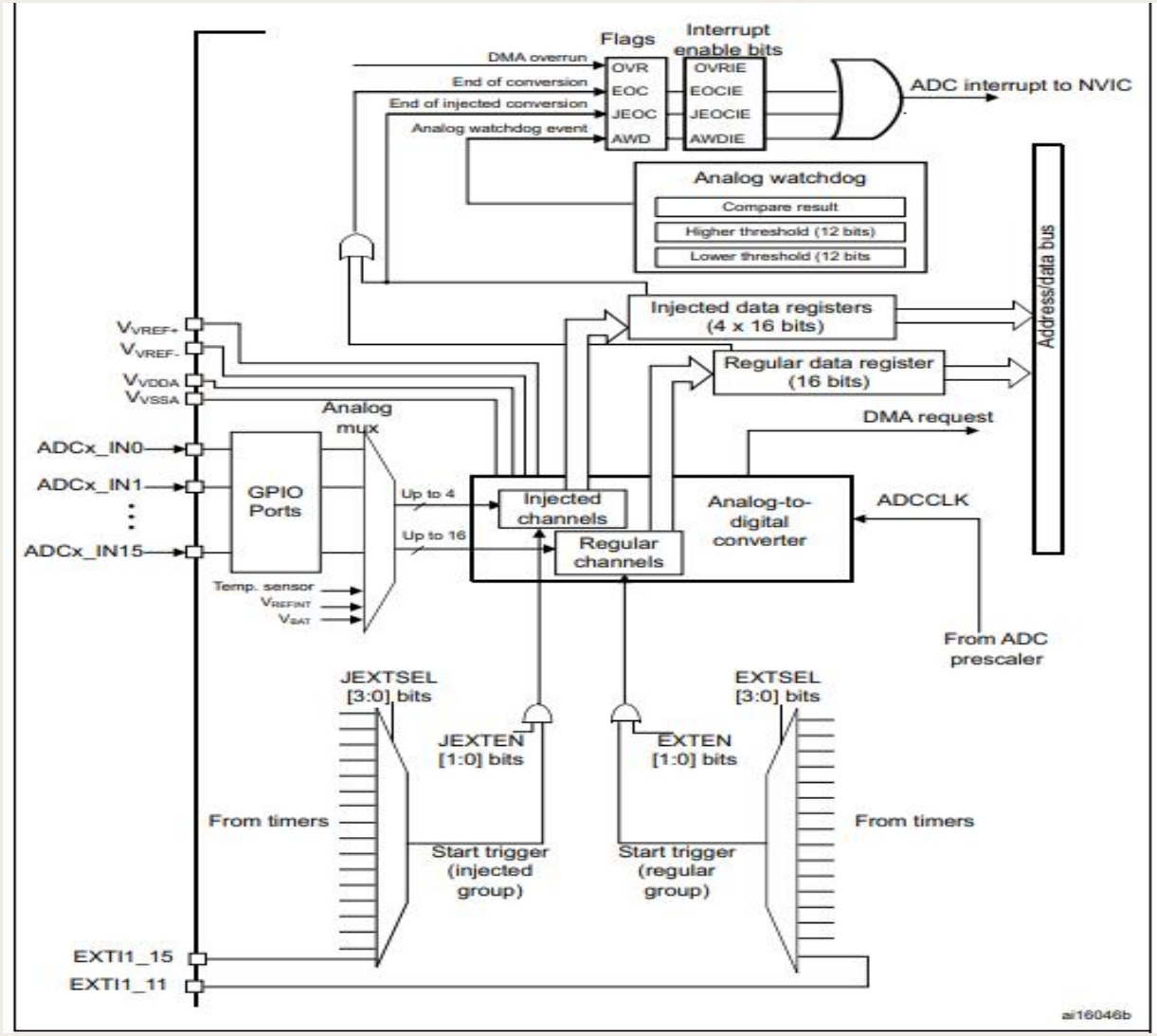
ADC introduction

- The 12-bit ADC is a successive approximation analog-to-digital converter. It has up to 19 multiplexed channels allowing it to measure signals from 16 external sources, two internal sources, and the VBAT channel. The A/D conversion of the channels can be performed in single, continuous, scan or discontinuous mode. The result of the ADC is stored into a leftor right-aligned 16-bit data register.

ADC main features

- STM32F446 ADC Features
- Three independent 12-bit ADCs
- Up to 24 external channels
- Resolution:
  - *12-bit*
  - *10-bit*
  - *8-bit*
  - *6-bit*
- Regular conversion
- Injected conversion
- Scan mode
- Continuous conversion
- Discontinuous mode
- DMA support
- Analog watchdog
- External trigger support

ADC Block Diagram



ADC Conversion Process

- Select ADC channel
- Configure sampling time
- Start conversion
- Sample analog voltage
- Convert to digital
- Store result in ADC_DR
- Read by CPU or DMA

ADC clock

The ADC features two clock schemes:

1.Clock for the analog circuitry: ADCCLK, common to all ADCs This clock is generated from the APB2 clock divided by a programmable prescaler that allows the ADC to work at fPCLK2/2, /4, /6 or /8. Refer to the datasheets for the maximum value of ADCCLK.

2Clock for the digital interface (used for registers read/write access) This clock is equal to the APB2 clock. The digital interface clock can be enabled/disabled individually for each ADC through the RCC APB2 peripheral clock enable register (RCC_APB2ENR).

Single conversion mode

- Once the conversion of the selected channel is complete:
- If a regular channel was converted:
  - The converted data are stored into the 16-bit ADC_DR register
  - The EOC (end of conversion) flag is set
  - An interrupt is generated if the EOCIE bit is set
- If an injected channel was converted:
  - The converted data are stored into the 16-bit ADC_JDR1 register
  - The JEOC (end of conversion injected) flag is set
  - An interrupt is generated if the JEOCIE bit is set Then the ADC stops.

Continuous conversion mode

- In continuous conversion mode, the ADC starts a new conversion as soon as it finishes one. This mode is started with the CONT bit at 1 either by external trigger or by setting the SWSTART bit in the ADC_CR2 register (for regular channels only).
- After each conversion:
    - If a regular group of channels was converted:
        - The last converted data are stored into the 16-bit ADC_DR register
        - The EOC (end of conversion) flag is set
        - An interrupt is generated if the EOCIE bit is set

Multi ADC mode

- In devices with two ADCs or more, the Dual (with two ADCs) and Triple (with three ADCs) ADC modes can be used
- The four possible modes below are implemented:
    - Injected simultaneous mode
    - Regular simultaneous mode
    - Interleaved mode
    - Alternate trigger mode

ADC registers

- ADC status register (ADC_SR)
- ADC control register 1 (ADC_CR1)
- ADC control register 2 (ADC_CR2)
- ADC sample time register 1 (ADC_SMPR1)
- ADC sample time register 2 (ADC_SMPR2)
- ADC injected channel data offset register x (ADC_JOFRx) (x=1..4)
- ADC watchdog higher threshold register (ADC_HTR)
- ADC watchdog lower threshold register (ADC_LTR)
- ADC regular sequence register 1 (ADC_SQR1)
- ADC regular sequence register 2 (ADC_SQR2)
- ADC regular sequence register 3 (ADC_SQR3)
- ADC injected sequence register (ADC_JSQR)
- ADC injected data register x (ADC_JDRx) (x= 1..4)
- ADC regular data register (ADC_DR)
- ADC Common status register (ADC_CSR)
- ADC common control register (ADC_CCR)

ADC Applications

- Temperature Monitoring
- Battery Voltage Measurement
- Medical Devices
- Robotics
- Industrial Automation
- IoT Sensors
- Motor Control
- Audio Processing

Advantages

- High accuracy
- Multiple channels
- DMA support
- Flexible trigger sources
- Multiple resolutions
- Fast conversion

THANK YOU